Highway Engineering Problem Session-1

Question 1: A truck is moving along a 8% upgraded road section with a constant speed of 40 km/h. The motor power of the truck is 170kW and the weight of the truck is 5 tons.

a) Find the traction force Ztr.
b) Find the amount of useful load that the truck can carry.
c) If this truck stops on the road, can it move again with the useful load calculated in part b.

Given: rolling resistance coefficient, μ_r=0.030; air resistance coefficient, K=0.070; F=0.9*b*h; b=2.30 m, h=2.70 m; $W_h = 0.75KFV^2$; motor efficiency coefficient, η=0.85, Sliding friction coefficient, *f*=0.60; The adherence weight, Q_m=2Q/3

Solution:

a) $Ztr = 3600 * \left(\frac{Nm}{V}\right) * \eta$ ➔➔ $Ztr = 3600 * \left(\frac{170}{40}\right) * 0.85$ ➔➔ $Ztr = 13005$ N

b) $Ztr = \sum W$; $\sum W = Wy + Wh + We$

Wy = μ_r * Q ➔➔ 0.030 * Q ➔➔ 0.03Q

$W_h = 0.75KFV^2$ ➔➔ $0.75*0.07*(0.9*2.30*2.70)*40^2$ ➔➔ 469.48 N

$We = \left(\frac{Q*s}{100}\right)$ ➔➔ $We = \left(\frac{Q*8}{100}\right)$ ➔➔ 0.08Q

$\sum W$ = 0.03Q + 0.08Q + 469.48 = 13005 ➔➔ Q = 113959.27 N ➔➔ 11395.93 kg

Useful load = 11395.93 Kg – 5000 Kg = 6395.93 Kg

c)

According to the two conditions of motion, for a stopping vehicle the following situations can be written:

1. $Z_{tr} < \Sigma W < Q_m.f$ → The vehicle does not move (no motion)

2. $\Sigma W < Z_{tr} < Q_m.f$ → The vehicle starts to move with the normal motion (rolling condition)

3. $\Sigma W < Q_m.f < Z_{tr}$ → The vehicle starts to move while spinning (forward motion with spinning)

4. $Q_m.f < \Sigma W < Z_{tr}$ → The vehicle spins and does not move (spinning without motion)

$Q_m * f > Z_{tr}$ (required for starting the motion)

$Q_{m*} f = (2/3) * 113959.27 * 0.6 = 45583.71\ N$

$Z_{tr} = \sum W = Wy + Wh + We$

Wh=0 vehicle is not moving

Wy = μ_r * Q ➔➔ 0.030 * 113959.27 = 3418.78 N

We=0.08 * Q ➔➔ 0.08 * 113959.27 = 9116.74 N

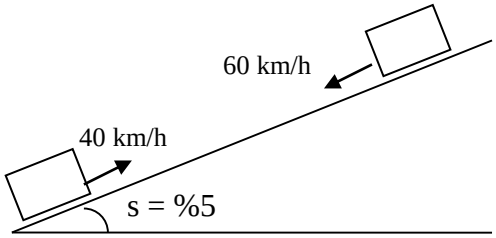
$\sum$W =12535.52 N

45583.71 N > 12535.52 N so it can start moving again.

Question 2: A vehicle is moving along a 5% downgraded one-way road section with a constant speed of 60km/h. At the same time another vehicle has entered this road section in the opposite direction by mistake with a constant speed of 40 km/h. What should be the minimum sight distance in order to prevent the collision ?

Given: Sliding friction coefficient, f = 0.60; Perception-reaction time (sec) t_r = 1 sec (for both vehicles)

Solution:



s = -%5
V_1 = 60 km/h
V_2 = 40 km/h

brake safety distance=l_{fe}

$$l_{fe} = 0.278 * V * t_r + 0.00394 * \frac{V^2}{f \mp s}$$

$$l_{fe2} = 0.278 * 40 * 1 + 0.00394 * \frac{40^2}{0.6 + 0.05} = 20.82m$$

$$l_{fe1} = 0.278 * 60 * 1 + 0.00394 * \frac{60^2}{0.6 - 0.05} = 42.47m$$

$$l_{\min} = l_{fe1} + l_{fe2} = 20.82 + 42.47 = 63.29m$$

Question 3: A car driver is moving on a slopeless road section with a constant speed of 90 km/h in a rainy day. The driver sees a pedestrian is crossing the road approximately 100 meters ahead.

a) Check if the car hits the pedestrian.

b) If the car hit the pedestrian, what should have been the maximum speed of the car in order to prevent collision?

Given: Sliding friction coefficient, f = 0.35; Perception-reaction time (sec) t_r = 0,75 s

Solution:

a)

$$l_{fe} = 0.278 * V * t_r + 0.00394 \frac{V^2}{f \mp s} \rightarrow\rightarrow l_{fe} = 0.278 * 90 * 0.75 + 0.00394 \frac{90^2}{0.35 \mp 0}$$

= 109.95 m > 100 m hits the pedestrian !!!

b)

In order to prevent collision it should stop within 100 meters.

l_{fe} <= 100 m

$$100 = 0.278 * V * t_r + 0.00394 \frac{V^2}{f \mp s} \quad \rightarrow\rightarrow 100 = 0.278 * V * 0.75 + 0.00394 \frac{V^2}{0.35}$$

$$\rightarrow\rightarrow 0.01126*V^2 + 0.2085*V - 100 = 0$$

Solve the equation and max speed;

V_1= 85.50 km/h

Note = $ax^2 + bx + c$ roots of the equation$\rightarrow\rightarrow$ $X_{1,2} = \frac{-b \mp \sqrt{b^2 - 4ac}}{2a}$

Question 4: A car moving on a slopeless road with a constant speed of 90 km/h. This car wants to pass the truck moving in front of the car with a constant speed of 60 km/h.

a) What should be the required distance for the car to pass the truck safely? And what is the total time period for passing? ($d_i = 8 + 0.3.V_i$)

b) In case of a car coming from the opposite direction with a constant speed of 70 km/h what is the required passing-sight distance?

c) In case of both of the vehicles are moving a speed of 60 km/h and the car in the back passing the front vehicle by accelerating from 60 km/h to 90km/h in 6 seconds. What should be passing time and distance? Perception-reaction time (sec) t_r = 1 sec

d) In case of both of the vehicles are moving a speed of 60 km/h and the vehicle in the back passing the front vehicle by accelerating from 60 km/h to 90km/h in 6 seconds. And a vehicle is coming from the opposite direction with a constant speed of 70 km/h what is the required passing-sight distance?

Solution:

a)

$d_1=8+0.3V_1$ ➔➔ 8 + 0.3*90 = 35 m ; $d_2=8+0.3V_2$ ➔➔ 8 + 0.3*60 = 26 m

$$ts = \frac{d_1 + d_2}{V_1 - V_2} \quad \rightarrow\rightarrow \quad ts = \frac{35 + 26}{(90 - 60) * \frac{1}{3.6}} \quad \rightarrow\rightarrow \quad \text{ts} = 7.32 \text{ sn}$$

$ls=ts*V_1$ ➔➔ $$ls = \frac{d_1 + d_2}{V_1 - V_2} * V_1 \quad \rightarrow\rightarrow \quad ls = \frac{35 + 26}{(90 - 60)} * 90 = 183 \text{ m}$$

b)

$ls=ts\ (V_1+V_3)$ ➔➔ $$ls = \frac{d_1 + d_2}{V_1 - V_2} * (V_1 + V_3) \quad \rightarrow\rightarrow \quad \frac{35 + 26}{90 - 60} * (90 + 70) = 325.33 \text{ m}$$

c)

$d_1= d_2 = 26$ m

$t_s=t_r+[2(d/a)^{0.5}]$ ➔➔ $t_s=1+[2(26/1.39)^{0.5}] = 9.65$ sn

$$a = \frac{\Delta V}{\Delta t} \quad \rightarrow\rightarrow \quad \gamma = \frac{(90 - 60)}{6} * \frac{1}{3.6} = 1.39 \text{ m/sn}^2$$

$ls=ts*V_2 + 2d$ ➔➔ 9.65*(60/3.6) + 2*26 = 212,83 m

d)

$Ls=l_{si} + l_{sk}$ ➔➔ $Ls= ts*V_2 + 2d + V_3*ts$

$Ls=9.65*(60/3.6) + 2*26 + (70/3.6)*9.65$ ➔➔ Ls= 400.47 m

Question 5: A car moving on a slopeless road with a constant speed of 90 km/h. This car passed the bus moving in front of the car with a constant speed of 70 km/h. Just after the passing car driver noticed an obstacle on the road and pushed the break pedal. The car stopped 1 m before the obstacle.

a) At the beginning of the passing, what is the distance between the car and the obstacle?
b) Check if the bus hits the car during this period?

Given: $d_i=7+0.2V_i$; Sliding friction coefficient, f = 0.40; Perception-reaction time (sec) t_r = 2 sec (for both vehicles)

Solution: a)

$d_1=7+0.2V_1$ ➔➔ 7 + 0.2*90 = 25 m

$d_2=7+0.2V_2$ ➔➔ 7 + 0.2*70 = 21 m

$$ls = \frac{d_1 + d_2}{V_1 - V_2} * V_1 \quad \rightarrow\rightarrow \quad ls = \frac{25 + 21}{90 - 70} * 90 \quad \rightarrow\rightarrow \quad \text{ls} = 207 \text{ m}$$

$$l_{fe} = 0.278 * V_1 * t_r + 0.00394 \frac{V_1^2}{f \mp s} \rightarrow\rightarrow l_{fe} = 0.278 * 90 * 2 + 0.00394 \frac{90^2}{0.4} = 129.83\text{m}$$

$l = l_s + l_{fe} + 1 \rightarrow\rightarrow 207 + 129.83 + 1 = 337.83$ m

b) $l_{fe} = (0.278 * 70 * 2) + (0.278 * 70 * 2) + 0.00394 \frac{70^2}{0.4} = 126.11$ m

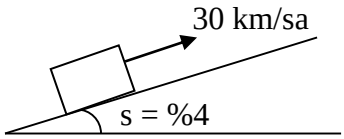
$l_f = 129.83$ m (automobile)

126.11 < 129.83+21 Therefore, no crash occurs.

Question 6: A loaded truck with a weight of 12 tons, is moving on a 4% upgraded road section with a constant speed of 30 km/h . What should be its motor power (kW) in order to keep moving with constant speed?

Given: rolling resistance coefficient $\mu_r = 0.025$; air resistance coefficient, K = 0.060; F = 0.9*b*h; b = 2.20 m , h = 2.50 m; $W_h = 0.75KFV^2$; motor efficiency coefficient, $\eta = 0.85$.

Solution:



m = 12 ton
N_m = ?

The required condition for moving with constant speed $Z_{tr} = \Sigma W$
$\Sigma W = W_y + W_h + W_e$

We : Grade resistance
Wy : rolling resistance
Wh : Air resistance
Q : Weight of the vehicle

$W_y = \mu_r * Q = \mu_r * m * g = 0.025 * 12000 * 10 = 3000$ N

$W_h = 0.75.K.F.V^2 = 0.75 * K * (0.9 * b * h) * V^2$
$W_h = 0.75 * 0.060 * (0.9 * 2.20 * 2.50) * 30^2 = 200.48$ N

$W_e = Q * s = m * g * s = 12000 * 10 * 0.04 = 4800$ N

$\Sigma W = W_y + W_h + W_e = 3000 + 200.48 + 4800 = 8000.48$ N

$$N_m = Z_{tr} . \frac{V}{3600 . \eta}$$

$$N_m = \frac{8000.48 * 30}{3600 * 0.85} = 78.44 kW$$

Question 7: A vehicle with a weight of 1050 kg has a Motor Power value of 70 Hp. The dimensions of this vehicle are 1,50 m width and 1,70 m height. Calculate the upgrade level that this vehicle can move with a constant speed of 120 km/h?

Given: rolling resistance coefficient = 0.020 N/N
air resistance coefficient= 0.025
motor efficiency coefficient = 0.75

Solution:

Q = 1050 kg = 10500 N

Horsepower to Kilowatts ➔ N_m = 70 BB = 70 * 0.735 = 51.45 kW

$$Z_{tr} = 3600 * \frac{N_m}{V} * \eta = 3600 * \frac{51.45}{120} * 0.75 = 1157.625 \text{ N}$$

$\sum W = W_y + W_h + W_e$

$W_y = \mu_r * Q = 0.020 * 10500 = 210$ N

$W_h = 0.75 * K * F * V^2 = 0.75 * 0.025 * (0.9 * 1.50 * 1.70) * 120^2 = 619.65$ N

$$W_e = \frac{Q * s}{100} \qquad Z_{tr} = \sum W$$

$1157.625 = 210 + 619.65 + W_e$

$W_e = 327.975$ N

$$W_e = \frac{10500 .s}{100} = 327.975 \quad => s = 3.12\%$$

Question 8: A vehicle is moving with a constant speed of 70 km/h along a road section with a sliding friction coefficient value of 0,50. Calculate the brake safety distances for the cases that the road section is 5% downgraded and 5% upgraded. (Perception-reaction time (sec) t_r = 0.75 sn)

Solution:

$l_{fe} = l_r + l_f$

$$l_f = 0.00394 * \frac{V^2}{f \mp s}$$

$l_r = 0.278 * V * t_r$

$$l_{fe} = 0.278 * V * t_r + 0.00394 * \frac{V^2}{f \mp s}$$

If upgraded

$$l_{fe} = 0.278 * V * t_r + 0.00394 * \frac{V^2}{f + s}$$

$$l_{fe} = 0.278 * 70 * 0.75 + 0.00394 * \frac{70^2}{0.50 + 0.05} = 14.595 + 35.10 = 49.695 \text{ m.}$$

If downgraded:

$$l_{fe} = 0.278 * V * t_r + 0.00394 * \frac{V^2}{f - s}$$

$$l_{fe} = 0.278 * 70 * 0.75 + 0.00394 * \frac{70^2}{0{,}50 - 0{,}05} = 14.595 + 42.90 = 57.495 \text{ m.}$$

Question 9: A bus is moving along a straight road section with a constant speed of 50 km/h. A car moving behind the bus wants to pass the bus by accelerating from 50 km/h to 90 km/h in 7 seconds. What is the time period and distance for safe passing? (t_r = 1 s, d = 7 + 0.3V)

Solution:

7 sec (90-50)*(1/3.6)
1 sec *a*

a = (90-50) /(3.6*7) = 1,59 m/sec^2

d = 7 + 0.3*50 = 22 m $t_s = t_r + [2(d/a)^{0,5}] = 1 + [2(22/1.59)^{0,5}] = 8.44$ sec

$$l_{si} = V_2 * t_{si} + 2 * d = \frac{50}{3.6} * 8.44 + 2 * 22 = 161.22 m$$